Completed

FIFO Page Replacement Algorithm

Write a C program to implement the FIFO Page Replacement Algorithm. The program should take the number of frames and a reference string (sequence of page requests) as input. For each page in the reference string, determine if it results in a 'Page Hit' or a 'Page Fault'. Display the state of the frames after each request and calculate the total number of page faults and hits.

Input Format

1. The first line of input contains an integer representing the number of pages in the reference string.
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The total number of Page Faults and Page Hits are displayed

Select Language : C (GCC 9.2.0)

```c
#include <stdio.h>
int main() {
    int pages, frames;

    printf("Enter number of pages in reference string: \n");
    scanf("%d", &pages);

    int ref[pages];
    printf("Enter the reference string: \n");
    for (int i = 0; i < pages; i++) {
        scanf("%d", &ref[i]);
    }

    printf("Enter number of frames: \n");
    scanf("%d", &frames);

    int frame[frames];
    int front = 0;
    int faults = 0, hits = 0;

    // Initialize frames
    for (int i = 0; i < frames; i++) {
        frame[i] = -1;
```

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Input Format

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Output Format

The total number of Page Faults and Page Hits are displayed

Select Language : C (GCC 9.2.0)

```c

        // Check for hit
        for (int j = 0; j < frames; j++) {
            if (frame[j] == ref[i]) {
                found = 1;
                hits++;
                break;
            }
        }

        // Page fault
        if (!found) {
            frame[front] = ref[i];
            front = (front + 1) % frames;
            faults++;
        }
    }

    printf("\nTotal Page Faults: %d\n", faults);
    printf("Total Page Hits: %d\n", hits);

    return 0;
}
```

Input :

```
7
1 3 0 3 5 6 3
3
```

Expected Output :

```
Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 6
Total Page Hits: 1
```

Output :

```
Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 6
Total Page Hits: 1
```

Completed

LRU Page Replacement Algorithm

Write a C program to implement the LRU Page Replacement Algorithm. The program should accept a reference string and the number of frames. For each page request, identify whether it results in a Hit or a Fault. If a Fault occurs and the frames are full, replace the page that was accessed least recently. Display the frame status at each step and calculate the total number of Page Faults and Hits

Input Format

1. The first line of input contains an integer representing the number of pages in the reference string. \
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The output displays the total number of Page Faults and the total number of Page Hits after processing the entire reference string using the LRU page replacement algorithm.

Select Language : C (GCC 9.2.0)

```c
#include <stdio.h>

int main() {
    int pages, frames;

    printf("Enter number of pages in reference string: \n");
    scanf("%d", &pages);

    int ref[pages];
    printf("Enter the reference string: \n");
    for (int i = 0; i < pages; i++) {
        scanf("%d", &ref[i]);
    }

    printf("Enter number of frames: \n");
    scanf("%d", &frames);

    int frame[frames], time[frames];
    int faults = 0, hits = 0, counter = 0;

    // Initialize frames
    for (int i = 0; i < frames; i++) {
        frame[i] = -1;
```

Completed

LRU Page Replacement Algorithm

Write a C program to implement the LRU Page Replacement Algorithm. The program should accept a reference string and the number of frames. For each page request, identify whether it results in a Hit or a Fault. If a Fault occurs and the frames are full, replace the page that was accessed least recently. Display the frame status at each step and calculate the total number of Page Faults and Hits

Input Format

1. The first line of input contains an integer representing the number of pages in the reference string. \
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The output displays the total number of Page Faults and the total number of Page Hits after processing the entire reference string using the LRU page replacement algorithm.

Select Language : C (GCC 9.2.0)

```c
        // If not found → Page Fault
        if (!found) {
            int lru = 0;

            // Find least recently used
            for (int j = 1; j < frames; j++) {
                if (time[j] < time[lru]) {
                    lru = j;
                }
            }

            counter++;
            frame[lru] = ref[i];
            time[lru] = counter;
            faults++;
        }
    }

    printf("\nTotal Page Faults: %d\n", faults);
    printf("Total Page Hits: %d\n", hits);

    return 0;
}
```

Input :

```
5
1 2 3 4 5
5
```

Expected Output :

```
Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 5
Total Page Hits: 0
```

Output :

```
Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 5
Total Page Hits: 0
```